Collapse / Resolution Framework (CRF)

A Computational Framework for Born-Aligned Selection, Post-Selection Dynamics, and Constraint Stabilization

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CRF is a constrained computational architecture for studying layered post-selection dynamics under Born-aligned first selection. It is not a collapse model and does not claim modification of quantum mechanics.

Author's Note:

This work represents the original research, analysis, and theoretical development of the author. Public versions are archived through DOI-backed repositories to maintain a transparent historical record of the framework's evolution.

Contents

Collapse / Resolution Framework (CRF) 1
A Computational Framework for Born-Aligned Selection, Post-Selection Dynamics, and Constraint Stabilization 1
Contents 1
Working Paper Notice 3
1. Abstract 3
2. Introduction 4
3. Core CRF Architecture 5
Layer 1 — Born Selection 5
Mechanism 5
Formula 5
Empirical Behavior 6
Interpretation 6

Invariant 1 — Born Alignment of First Selection 6
Layer 2 — Exposure / Dwell 6
Mechanism 6
Formula 6
Empirical Behavior 6
Interpretation 6
Layer 3 — Survival / Capture 7
Mechanism 7
Formula 7
Empirical Behavior 7
Interpretation 7
Layer 4 — Resolution / Born Fallback 7
Mechanism 7
Formula 7
Empirical Behavior 8
Interpretation 8
4. Operator Architecture 8
Operator Summary 8
5. Mathematical Formulation 10
4.1 Exponential Race 10
Lemma — Exact Born Selection 10
5.2 Detector Signal and Dwell 10
5.3 Survival / Capture 11
5.4 Reroute Rule 11
5.5 Born Fallback 11
5.6 Stabilized Survival 12
5.7 Stabilizer Properties 12
6. Parameter Map 13
7. Detector Placement 14
8. Empirical Tension / Missing Stabilizer 14
Observation 14
8.1 Recovery Scaling Observation 15
*Companion Structural Analogy Paper— Triple-Slit Response Stability 16

9. Conclusion 17
Appendix A — Dwell-Asymmetry Stabilizer 18
Raw Survival 18
Stabilized Survival 18
Appendix B — Gamma Selection Validation 19
Empirical Admissible Window 19
Validation Results 19
Closing Statement 20

Working Paper Notice

This document presents computational and structural investigations into post-selection dynamics within a constrained computational architecture referred to as Collapse / Resolution Framework (CRF).

CRF is an experimental computational framework designed to study how Born-aligned first selection behaves under asymmetric detector geometry, dwell-dependent survival dynamics, and stabilization constraints.

CRF is not an objective collapse model and does not propose replacement of the Born rule.

The framework does not claim violation of quantum mechanics, replacement of the Born rule, or discovery of new physical laws. All results remain bounded to the tested architecture and are presented as computational findings rather than physical proofs.

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1. Abstract

Collapse / Resolution Framework (CRF) is a constrained computational architecture for studying layered post-selection dynamics under Born-aligned first selection. The framework separates selection, dwell generation, survival behavior, and stabilization into distinct operational layers, allowing detector-geometry–dependent amplification to be analyzed without modifying the Born rule.

Across 36 detector configurations, first selection remains exactly Born-aligned, while downstream survival dynamics generate geometry-dependent amplification capable of producing substantial divergence from Born-like final statistics. A dwell-asymmetry stabilizer with coupling γ compresses runaway survival spread while preserving detector-dependent

structure within the tested architecture. Subsequent recovery-scaling results suggest that stabilization and recovery may be distinct downstream processes. CRF therefore closes as a computational architecture for studying admissibility, stabilization, and invariant-preserving dynamics under structured internal asymmetry. No quantum-mechanical claims are made.

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2. Introduction

The Collapse / Resolution Framework (CRF) examines a structural question that is usually assumed away: does exact Born-aligned first selection guarantee Born-consistent final statistics once internal detector dynamics are allowed to matter?

To expose this question cleanly, CRF separates the selection pipeline into four operational layers:

- Layer 1 — Born-aligned first selection
- Layer 2 — Exposure / dwell generation
- Layer 3 — Survival / Capture dynamics
- Layer 4 — Resolution / fallback stabilization

This layered architecture allows CRF to isolate where geometry-dependent amplification enters the system and how it interacts with an invariant first-selection rule. The framework does not modify quantum mechanics; instead, it evaluates whether structured internal asymmetry can generate systematic downstream divergence beneath exact Born alignment.

The computational environment for CRF is the omniflow simplex system

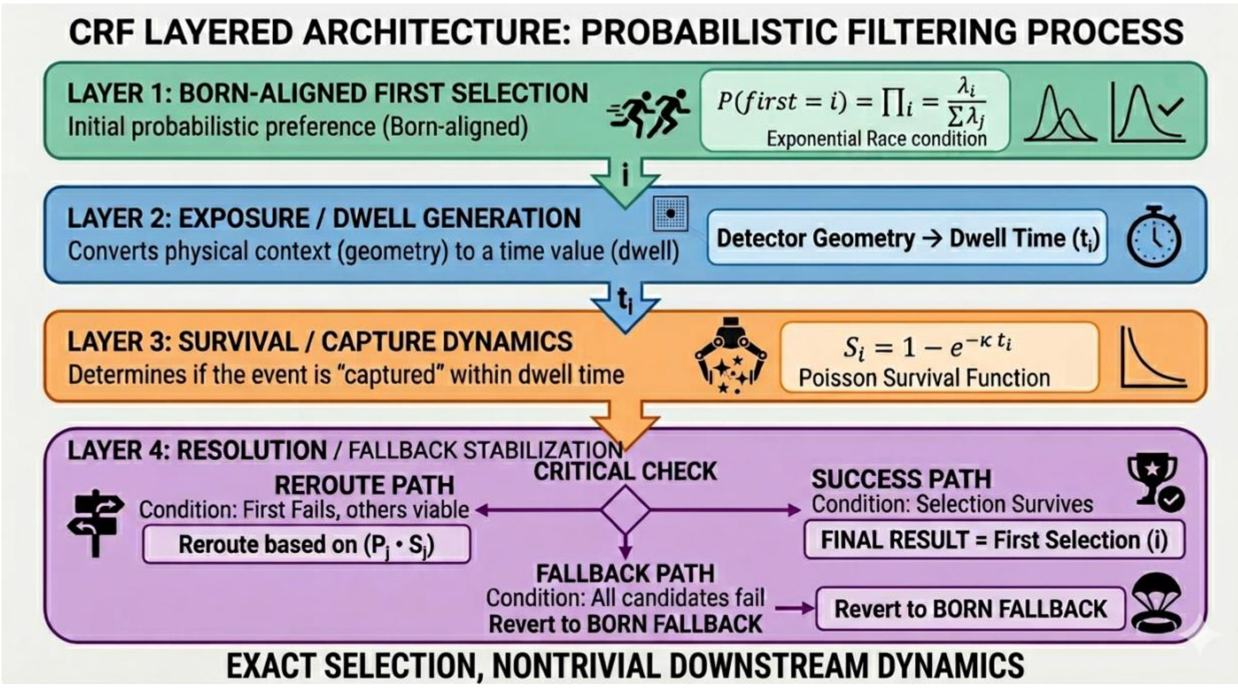
$$L + S + Q + D$$

with static bias field H acting on the 3-simplex. This symbolic structure is not itself a physical claim; it functions solely as the computational substrate supporting the layered architecture and validation environment.

Across this architecture, a consistent empirical tension emerges: symmetric detector geometries preserve Born-like behavior, while asymmetric geometries amplify dwell-dominant channels unless stabilization suppresses runaway asymmetry. CRF therefore studies whether admissibility constraints, stabilization behavior, or implicit calibration effects are required to maintain stable global statistical consistency under structured internal dynamics.

3. Core CRF Architecture

CRF is organized into four sequential operational layers.



Layer 1 — Born Selection

Mechanism

Each state participates in an exponential race process weighted by Born probabilities.

Formula

$$T_i \sim \mathrm{Exp}(\Lambda p_i)$$

Winner:

$$\mathrm{first} = \underset{i}{\mathrm{argmin}} T_i$$

Because the exponential distribution is memoryless:

$$P(\mathrm{first} = i) = p_i$$

exactly.

Empirical Behavior

Across all tested detector geometries and stabilization regimes, first-selection statistics remain Born-aligned.

Interpretation

The exponential race mechanism preserves exact Born-aligned first-selection probabilities across all tested regimes.

Invariant 1 — Born Alignment of First Selection

$$P(\text{first} = i) = p_i$$

The exponential race selector preserves exact Born-aligned first-selection probabilities independent of downstream survival dynamics, stabilization behavior, or detector asymmetry.

Layer 2 — Exposure / Dwell

Mechanism

Detector activation generates above-threshold exposure dwell after first selection.

Formula

Detector signal:

$$x_i(t) = A_i e^{-\max(t - T_i, 0)/\text{decay}_i} \mathbf{1}[t \geq T_i] + \epsilon_i(t)$$

Noise:

$$\epsilon_i(t) \sim \mathcal{N}(0, \sigma^2)$$

Threshold dwell:

$$\tau_i = \int_0^{T_{\text{total}}} \mathbf{1}\,[x_i(t) > \theta]\,dt$$

Empirical Behavior

Asymmetric detector geometry generates unequal dwell persistence across channels.

Interpretation

Exposure dwell converts Born-aligned first selection into kinetic detector persistence.

Layer 3 — Survival / Capture

Mechanism

Exposure dwell is converted into post-selection survival probability.

Formula

$$s_i = 1 - e^{-\kappa \tau_i}$$

with:

$$s_i \in [0,1]$$

Empirical Behavior

Symmetric detector geometry preserves Born-like behavior while asymmetric detector geometry amplifies dwell-dominant channels.

Interpretation

The Born rule governs first selection. The Survival / Capture layer governs whether that selection persists.

Layer 4 — Resolution / Born Fallback

Mechanism

Channels surviving the Survival / Capture layer proceed directly to final outcome resolution. Failed channels enter reroute resolution.

Formula

If:

$$u < s_{\text{first}}$$

then:

$$\text{final} = \text{first}$$

Otherwise reroute weights are formed:

$$w_j = p_j s_j$$

and sampled categorically.

Conditional reroute probability:

$$P(\text{final} = j \mid \text{fail}) = \frac{w_j}{\sum_k w_k}$$

Fallback condition:

$$\sum_j w_j = 0$$

triggers Born fallback resolution:

$$P(j) \propto p_j$$

Empirical Behavior

Fallback resolution eliminates unresolved outcomes while preserving stable global statistical behavior.

Interpretation

Resolution dynamics stabilize downstream selection behavior without modifying first-selection probabilities.

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4. Operator Architecture

The CRF pipeline separates into five operational operators:

Operator	Input	Output	Function
$\mathcal{D}$	(p, θ)	x	Competition geometry
τ	x	τ	Above-threshold dwell
s	τ	s	Raw survival dynamics
$\mathcal{P}_\gamma$	s	s^*	Dwell-asymmetry compression / stabilization
$\mathcal{R}$	s^*	outcome	Resolution / fallback selection

Operator Summary

The CRF pipeline is expressed as a sequence of operators acting on probability-bearing configuration data:

- **D — competition geometry** → detector signal
- **T — thresholding** → dwell
- **S — raw survival dynamics**

- **Pγ — dwell-asymmetry compression**
- **R — resolution / fallback selection**

The stabilizer Pγ acts as a bounded compression operator: it compresses survival spread toward the Born-weighted mean without renormalizing or erasing detector-dependent structure.

The stabilizer operator $\mathcal{P}_\gamma$ acts as a compression operator on survival asymmetry rather than a renormalization operator.

The stabilizer preserves ordering while suppressing runaway asymmetry.

Raw survival:

$$s_i = 1 - e^{-\kappa \tau_i}$$

Born-weighted mean survival:

$$\bar{s} = \sum_j p_j\, s_j$$

Dwell asymmetry:

$$A_\tau = 1 - \frac{\min_i(\tau_i + \epsilon)}{\max_i(\tau_i + \epsilon)}$$

Compression factor:

$$\lambda = \frac{1}{1 + \gamma A_\tau}$$

Stabilized survival:

$$s_i^* = \bar{s} + \lambda(s_i - \bar{s})$$

Full operator sequence:

$$(p, \theta) \xrightarrow{\mathcal{D}} x \xrightarrow{\tau} \tau \xrightarrow{s} s \xrightarrow{\mathcal{P}_\gamma} s^* \xrightarrow{\mathcal{R}} \text{outcome}$$

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5. Mathematical Formulation

4.1 Exponential Race

Lemma — Exact Born Selection

The exponential race selector yields exact Born-aligned first-selection probabilities.

Race dynamics:

$$T_i \sim \mathrm{Exp}(\Lambda p_i)$$

For exponential race variables with rates Λp_i, the probability that channel i wins the race is:

$$P(\text{first} = i) = \frac{\Lambda p_i}{\sum_k \Lambda\, p_k}$$

Since the Born probabilities satisfy:

$$\sum_k p_k = 1$$

the expression reduces to:

$$P(\text{first} = i) = p_i$$

Thus, first selection remains exactly Born-aligned.

5.2 Detector Signal and Dwell

Detector signal:

$$x_i(t) = A_i e^{-\max(t - T_i, 0)/\mathrm{decay}_i} \mathbf{1}[t \geq T_i] + \epsilon_i(t)$$

Noise:

$$\epsilon_i(t) \sim \mathcal{N}(0, \sigma^2)$$

Threshold dwell:

$$\tau_i = \int_0^{T_{\text{total}}} \mathbf{1}\,[x_i(t) > \theta]\, dt$$

The dwell variable τ_i measures the total time that detector channel i remains above threshold.

5.3 Survival / Capture

Raw survival dynamics:

$$s_i = 1 - e^{-\kappa\tau_i}$$

with:

$$s_i \in [0,1]$$

Longer dwell increases survival probability, while shorter dwell suppresses persistence.

The Survival / Capture layer therefore transforms exposure dwell into post-selection persistence.

5.4 Reroute Rule

If the first-selected channel does not survive, reroute weights are formed by combining Born probability with survival weight:

$$w_j = p_j s_j$$

Conditional reroute probability:

$$P(\text{final} = j \mid \text{fail}) = \frac{w_j}{\sum_k w_k}$$

This rule allows survival structure to influence final outcomes without modifying the original Born-aligned first-selection process.

5.5 Born Fallback

If all reroute weights vanish:

$$\sum_j w_j = 0$$

then the system falls back to Born-weighted resolution:

$$P(j) \propto p_j$$

Since:

$$\sum_j p_j = 1$$

the fallback distribution is:

$$P(j) = p_j$$

Born fallback eliminates unresolved outcomes without changing the Survival / Capture formula.

5.6 Stabilized Survival

The stabilizer introduces bounded compression of survival asymmetry.

Born-weighted mean survival:

$$\bar{s} = \sum_j p_j \, s_j$$

Dwell asymmetry:

$$A_\tau = 1 - \frac{\min_i(\tau_i + \epsilon)}{\max_i(\tau_i + \epsilon)}$$

Compression factor:

$$\lambda = \frac{1}{1 + \gamma A_\tau}$$

Stabilized survival:

$$s_i^* = \bar{s} + \lambda(s_i - \bar{s})$$

The stabilizer does not erase detector-dependent structure. It compresses excessive survival spread toward the Born-weighted mean.

5.7 Stabilizer Properties

If no stabilizer is applied:

$$\gamma = 0$$

then:

$$\lambda = 1$$

and:

$$s_i^* = s_i$$

If dwell asymmetry vanishes:

$$A_\tau = 0$$

then:

$$\lambda = 1$$

and:

$$s_i^* = s_i$$

If stabilizer strength becomes large:

$$\gamma \to \infty$$

then:

$$\lambda \to 0$$

and:

$$s_i^* \to \bar{s}$$

Thus, $\mathcal{P}_\gamma$ acts as a compression operator, not a renormalization operator.

It suppresses runaway dwell asymmetry while preserving the structure of the CRF pipeline.

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6. Parameter Map

A total of 36 detector configurations were evaluated across coupled parameter sweeps:

Parameter	Sweep Values
κ	$0.3,\ 1.0,\ 3.0$
θ	$0.15,\ 0.25,\ 0.35$
Decay geometry	symmetric → asymmetric

Across the tested architecture, the dominant control axis is **decay asymmetry**.

Symmetric decay geometry reliably preserves Born-like behavior, while increasing asymmetry generates progressively larger divergence from Born alignment. High-κ lock-in regimes suppress this divergence and restore stable Born-like statistical structure.

This suggests that dwell amplification is governed primarily by **exposure asymmetry**, rather than instability in the first-selection mechanism.

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7. Detector Placement

The detector labels used throughout CRF (PD, APD, SNSPD, PMT, TIF, SQR) function as shorthand identifiers for abstract computational geometry regimes inspired by broad detector behavior classes. They are not intended as physically calibrated detector simulations and should not be interpreted as quantitative models of experimental hardware.

Representative detector placements within the parameter map are summarized below:

Detector	Region	L_1
PD	Born-like	0.0135
APD	Born-like	0.0233
SNSPD	Born-like	0.0065
PMT	Moderate divergence	0.0998
TIF	Strong divergence	0.2503
SQR	Strong divergence	0.2201

PD, APD, and SNSPD configurations naturally cluster in near-symmetric Born-like regions.

PMT configurations exhibit moderate asymmetry amplification while remaining partially stabilized.

TIF and SQR configurations occupy strongly asymmetric regions associated with maximal divergence from Born alignment.

Structurally, near-symmetric decay combined with high-κ stabilization naturally restores Born-like behavior. Strong asymmetry combined with moderate κ generates persistent survival amplification and large statistical divergence.

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8. Empirical Tension / Missing Stabilizer

Observation

CRF predicts systematic divergence from Born alignment under asymmetric detector geometry unless stabilization behavior suppresses runaway dwell amplification.

The central empirical tension of the framework is straightforward:

The Survival / Capture layer possesses no intrinsic calibration mechanism.

Under asymmetric geometry, dwell-dominant channels accumulate disproportionate survival amplification, generating persistent divergence from Born-like outcome statistics.

Within the tested architecture, three operational resolutions remain possible:

1. Real detector systems may already be physically calibrated into Born-like operating regions.
2. Additional stabilization structure may exist.
3. The detector model itself may be incomplete.

No resolution is claimed.

Later recovery-scaling tests suggest that stabilization and recovery should be treated separately. Stabilization suppresses dwell-induced asymmetry inside the Survival / Capture layer, while recovery acts after statistical drift has already appeared in the output distribution.

This distinction motivated subsequent recovery-scaling investigations described in Section 8.1.

8.1 Recovery Scaling Observation

Subsequent investigations explored whether detector-induced divergence exhibits a systematic recovery relationship rather than merely responding to fixed stabilization strength.

Using actual pre-stabilizer detector output vectors obtained from the CRF architecture, a simplex-constrained recovery operator was applied downstream of the Survival / Capture layer. The operator preserved normalization and positivity while reducing divergence from the Born reference distribution.

The recovery operator applies a bounded simplex-preserving contraction toward a supplied Born reference distribution while preserving normalization and positivity.

Across 90 independent detector/seed configurations (10 seeds × 9 detector classes), recovery magnitude exhibited an exceptionally strong linear relationship with initial deviation magnitude:

$r = 0.9994$

$R^2 = 0.9988$

Recovery_L1 ≈ 0.588 × Raw_L1

This result should not be interpreted as a derivation of Born statistics. It indicates proportional recovery toward a supplied Born reference under the tested operator.

The relationship remained stable across Born-like, asymmetric, and pathological detector classes.

Importantly, the result does not demonstrate derivation of the Born distribution. The operator was supplied with a Born reference distribution and acts as a recovery mechanism rather than a generative mechanism. The observation instead suggests that larger departures from Born alignment experience proportionally larger corrective pressure within the tested architecture.

This behavior is consistent with a drift-sensitive stabilization process:

small deviation → small recovery

large deviation → large recovery

Within the CRF framework, the result motivates a distinction between stabilization and recovery. The γ-stabilizer compresses dwell-induced asymmetry within the Survival / Capture layer, whereas the recovery operator acts on the resulting statistical distribution itself. The two mechanisms therefore occupy different locations within the layered architecture.

No physical interpretation is claimed. The result remains bounded to the tested computational framework. However, the observed proportional scaling provides evidence that recovery behavior may represent a distinct structural phenomenon worthy of separate investigation alongside dwell-asymmetry stabilization.

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*Companion Structural Analogy Paper— Triple-Slit Response Stability

A separate companion framework, *Triple-Slit Response Stability* (TSRS), explores an analogous layered architecture under exact Sorkin cancellation. In that framework, amplitudes remain unchanged while observable residual structure emerges entirely from configuration-dependent response dynamics, stabilization, recovery, hysteresis, and fatigue effects.

Importantly, the protected amplitude-level invariant remains exact throughout all tested regimes.

The companion framework therefore illustrates a structurally similar phenomenon to CRF:

- internal response dynamics may become highly structured,
- observable residual behavior may emerge,

- stabilization layers may suppress runaway asymmetry,
- while the deeper invariant remains preserved.

TSRS does not validate CRF and makes no quantum-mechanical claims. Its relevance here is architectural: both frameworks independently exhibit layered dynamics in which observable structure coexists with preserved foundational constraints.

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9. Conclusion

The Collapse / Resolution Framework (CRF) shows that exact Born-aligned first selection can coexist with highly structured downstream dynamics, including dwell asymmetry, survival amplification, and stabilization behavior, within a constrained computational architecture. Across all tested regimes, the exponential-race selector preserves exact Born-aligned first-selection probabilities, establishing a protected invariant that remains stable regardless of detector geometry, dwell conditions, or stabilization strength.

Downstream of this invariant, the Survival / Capture layer introduces geometry-dependent amplification that can substantially reshape final outcome statistics. Under symmetric detector geometry, dwell persistence remains balanced and survival probabilities track Born-like behavior. Under asymmetric geometry, however, dwell-dominant channels accumulate disproportionate survival weight, generating persistent divergence from Born-like final statistics even though first selection remains exactly Born-aligned. This divergence is not a numerical artifact: it is a structural consequence of allowing geometry-dependent dwell to feed into survival dynamics without intrinsic calibration.

Across the full parameter sweep, Born-like statistical behavior emerges most reliably under three conditions: **(1)** symmetric detector geometry, which minimizes dwell asymmetry; **(2)** high-κ lock-in, which suppresses survival spread; and **(3)** stabilization regimes that compress runaway dwell asymmetry without erasing detector-dependent structure.

The stabilizer behaves not as a universal correction but as an **admissibility constraint**. It preserves ordering, compresses excessive survival spread toward the Born-weighted mean, and prevents runaway amplification while allowing rich internal dynamics to remain visible beneath it. In this sense, stabilization functions as a structural boundary condition rather than a modification of the underlying selection rule.

CRF therefore closes not as a replacement for the Born rule, nor as a proposal for new physical dynamics, but as a constrained computational architecture for studying how stabilization, admissibility, detector asymmetry, and layered response behavior interact downstream of an invariant first-selection mechanism. The framework demonstrates that exact Born alignment at Layer 1 does not, by itself, guarantee Born-like final statistics once Layers 2–4 introduce geometry-dependent persistence and amplification.

The companion Triple-Slit Response Stability (TSRS) framework independently exhibits a related structural pattern: response-layer residual behavior may emerge while deeper invariants remain exact. In both architectures, the protected invariant is untouched, while layered internal dynamics generate configuration-dependent structure that must be stabilized to maintain global consistency.

Together, CRF and TSRS motivate further study of layered admissibility, stabilization behavior, and invariant-preserving dynamics within constrained computational architectures. They suggest that internal asymmetry, dwell persistence, and response-layer structure may play a larger role in downstream statistical behavior than is typically assumed, even when foundational invariants remain fully intact.

Appendix A — Dwell-Asymmetry Stabilizer

Raw Survival

$$s_i = 1 - e^{-\kappa\tau_i}$$

Stabilized Survival

The Born-weighted mean survival is defined as:

$$\bar{s} = \sum_j p_j\, s_j$$

Dwell asymmetry is defined as:

$$A_\tau = 1 - \frac{\min_i(\tau_i + \epsilon)}{\max_i(\tau_i + \epsilon)}$$

The compression factor is:

$$\lambda = \frac{1}{1 + \gamma A_\tau}$$

The stabilized survival value is then:

$$s_i^* = \bar{s} + \lambda(s_i - \bar{s})$$

This construction has the following limiting properties:

$\gamma = 0 \Rightarrow s_i^* = s_i\, A_\tau = 0 \Rightarrow s_i^* = s_i\, \gamma \to \infty \Rightarrow s_i^* \to \bar{s}$

The observed working stabilization regime in this architecture is:

$$\gamma \approx 0.75$$

The stabilizer suppresses runaway dwell asymmetry while preserving survival ordering and overall CRF structure.

Within the tested architecture, $\gamma \approx 0.75$ appears as the first observed interior stabilization point within the tested sweep.

Appendix B — Gamma Selection Validation

Empirical Admissible Window

$$\gamma_{\min} \in (0.63,\ 0.70]$$

The working value is:

$$\gamma_{\text{work}} \approx 0.75$$

The selection criteria were:

1. TIF and SQR divergence reduced below threshold,
2. Born-like detectors remain Born-like,
3. No collapse toward uniformity occurs.

Thus,

$$\gamma \approx 0.75$$

appears as the first observed interior stabilization point within the tested sweep.

Validation Results

Detector	L_1 raw	L_1 stabilized	ΔL_1
PD	0.0135	0.0257	-0.0122
APD	0.0219	0.0223	-0.0005
SNSPD	0.0059	0.0297	-0.0238
TIF	0.2424	0.1245	$+0.1179$
SQR	0.2228	0.1195	$+0.1033$
s35_worst	0.3577	0.1723	$+0.1854$
EXT_AS	0.2143	0.1130	$+0.1013$
EXT_PX	0.3923	0.1862	$+0.2060$

The stabilizer exhibits a measurable tradeoff: strongly asymmetric configurations experience substantial divergence reduction, while already Born-like configurations may experience small increases in deviation. This behavior reflects bounded asymmetry compression rather than universal error minimization.

Born-like detector configurations naturally remain stable under constrained stabilization.

Strongly asymmetric geometries generate substantial divergence from Born alignment.

Stabilization suppresses divergence without erasing detector-dependent structure.

The stabilizer therefore behaves as an **admissibility constraint** rather than a universal correction.

Closing Statement

All results presented here remain experimental and bounded to the tested architecture.

CRF does not demonstrate violation of the Born rule. Instead, the framework suggests that strong internal asymmetry, dwell dynamics, and response-layer structure may coexist with remarkably stable global statistical behavior.

The companion Triple-Slit Response Stability framework independently exhibits a related structural pattern in which observable residual behavior emerges while deeper invariants remain exact.

Together, the two frameworks motivate further study of layered admissibility, stabilization, detector asymmetry, and invariant-preserving dynamics within constrained computational architectures.

CRF remains a constrained computational architecture for studying post-selection dynamics under structured internal asymmetry.

All numerical outputs discussed in this paper are reproducible within the tested architecture using the stated detector configurations and parameter sweeps. Subsequent multi-seed validation produced qualitatively similar behavior across independent random initializations, indicating that the reported effects are not unique to the reference seed. Seed = 42 is retained throughout as the canonical reference run for consistency of presentation.